

軟體測試與資安檢測

教師：王凡

授課時間：每週三 9：10—12：00

博理館 113

課程代號：EE 5170

識別碼：921 U9550

民國 110 年春季班

QUIZ (2021/05/19)

Open-book, open-internet, no discussion

Name: _____ Department : _____ Student ID: _____

1. Considering the following context-free grammar of a simple programming language.

```
S ::= A; S
A ::= x=E | if (B) then { S } else { S } | while (B) {S}
B ::= E <= E | !B | B || B | B &&B
E ::= c | x | E + E | E-E | E*E | E/E
c is an integer constant and x is a string token.
```

Please write down a program to test a compiler for this programming language for terminal symbol coverage. You must explain why the program satisfies the terminal symbol coverage.

2. Continuing from problem 1, please write down another program to test a compiler for this programming language for production coverage. You must explain why the program satisfies the production coverage.

3. Continuing from problem 1, please write down yet another program to test a compiler for this programming language for derivation coverage. You must explain why the program satisfies the derivation coverage.

4. Continuing from problem 1, suppose we have three mutation operators: ABS and AOR. Please write a set of mutation requirements for mutation operator coverage.

5. Continuing from problem 4, please write down a mutant program for each requirement. Note that all mutant programs must be invalid according to the grammar so that they can be killed.